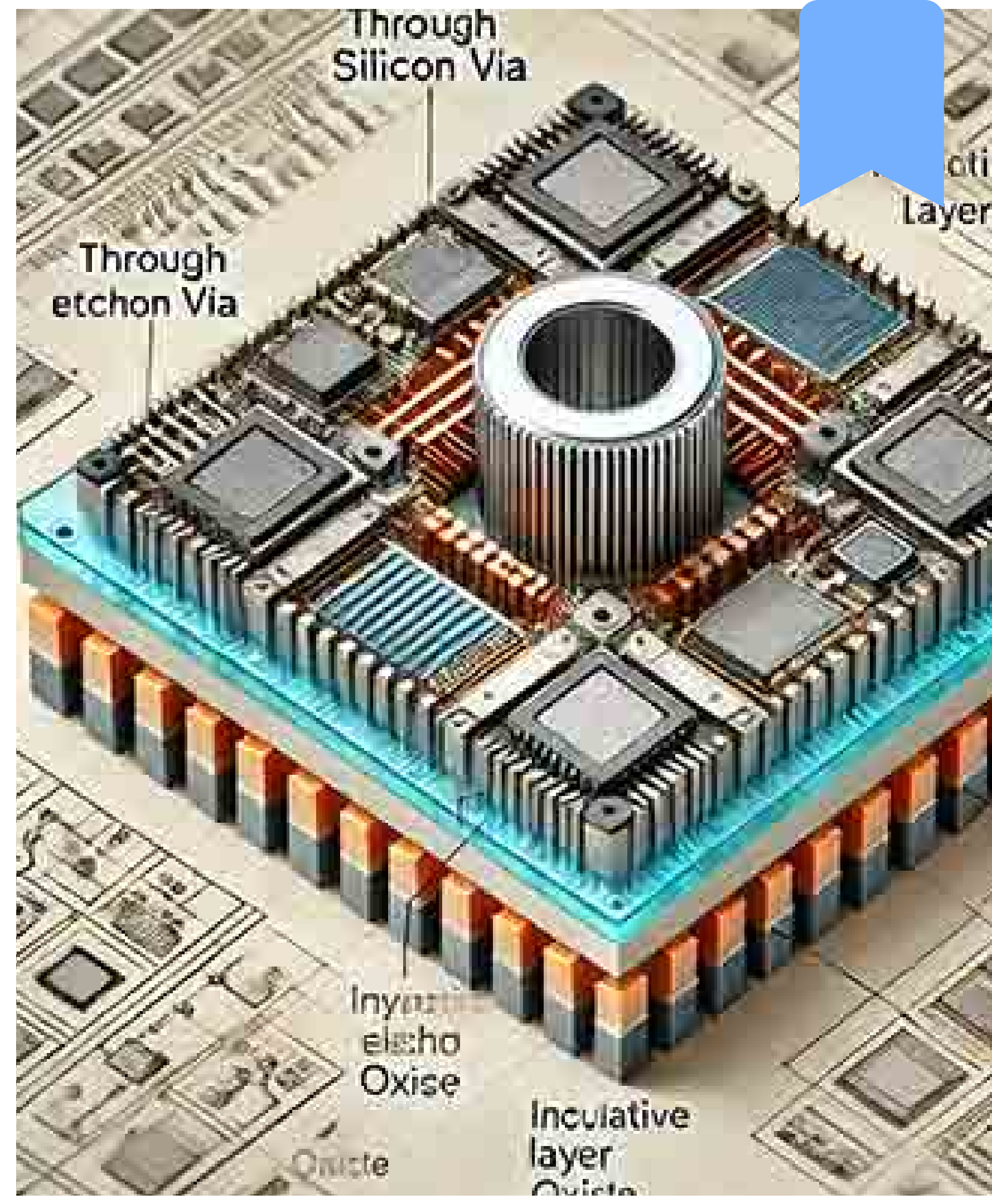


Comparative Study on Pinhole Defected GS TSV in Terms of Electrical Characterization.



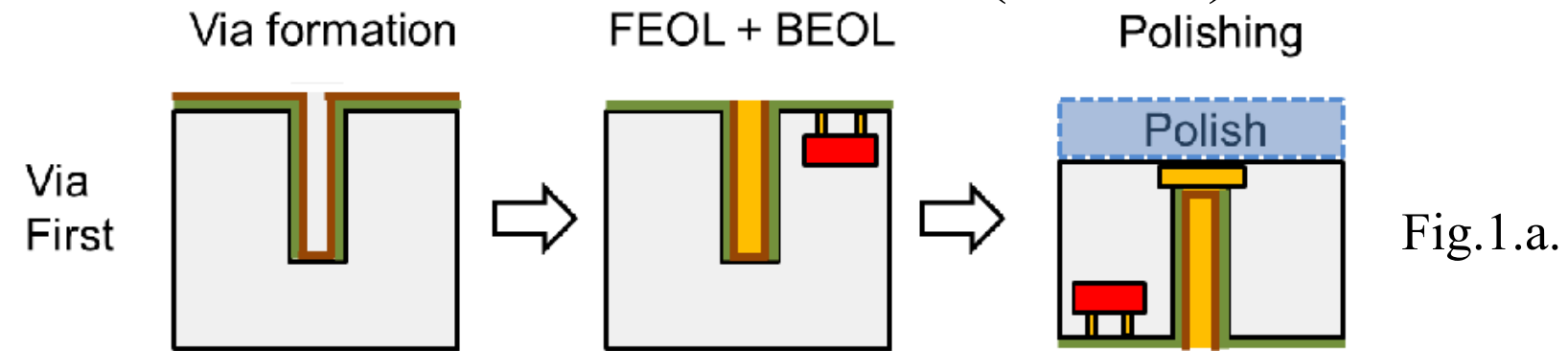
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-By K. Dhana Sri Keerthi

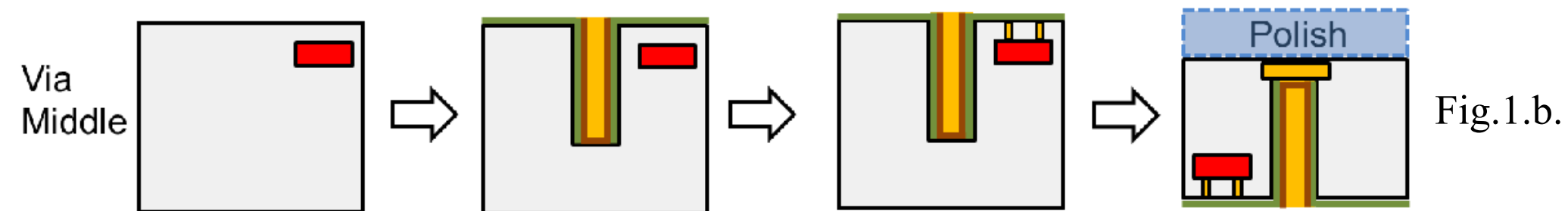


Via First, Middle & Last

➡ **Via First:** TSVs are formed before transistor (CMOS) fabrication.



➡ **Via Middle:** TSVs are formed after transistor fabrication but before metal interconnect layers.



➡ **Via Last:** TSVs are formed after the complete CMOS fabrication process.

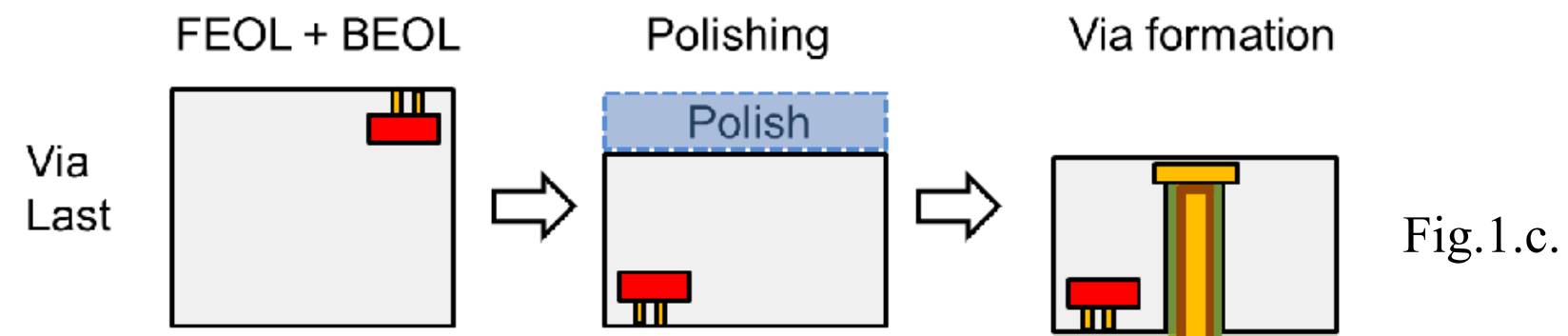


Fig.1 : a.Via First, 1.b. Via Middle, 1.c. Via Last

Why Choose Via Last for Fabrication?

TSV Type	Temperature Exposure	Reason
Via-First	High ($\sim 900\text{--}1100^{\circ}\text{C}$)	TSVs experience all CMOS processing temperatures after fabrication.
Via-Middle	Moderate ($\sim 300\text{--}500^{\circ}\text{C}$)	TSVs are formed after transistor fabrication, reducing thermal stress.
Via-Last	Low ($<400^{\circ}\text{C}$)	TSVs are fabricated after CMOS processing, minimizing thermal exposure.



- TSV fabrication occurs after CMOS transistor fabrication, so the transistors are not exposed to additional high temperatures.
- Reduces thermal stress and potential damage to devices.

Fabrication Work Flow

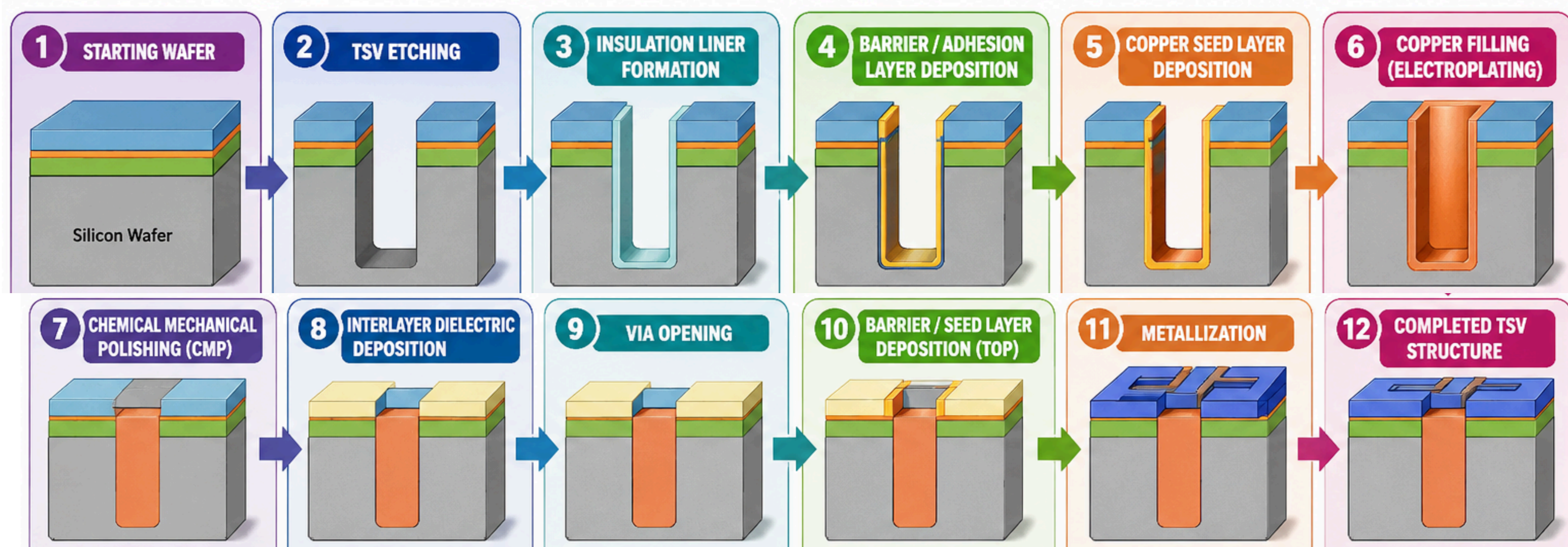


Fig.2. Via Fabrication Process.

Types of Defects

Void Defect: A void defect is defined as a tiny gap that forms inside the copper filled TSV during the filling process.

Delamination Defect: The Defect that occurs when the layers inside the chip start to crack due to Thermal Stress

Pinhole Defect: A pinhole defect is defined as a tiny hole in the insulator or barrier that causes leakage current to flow from TSV to the Si Substrate.

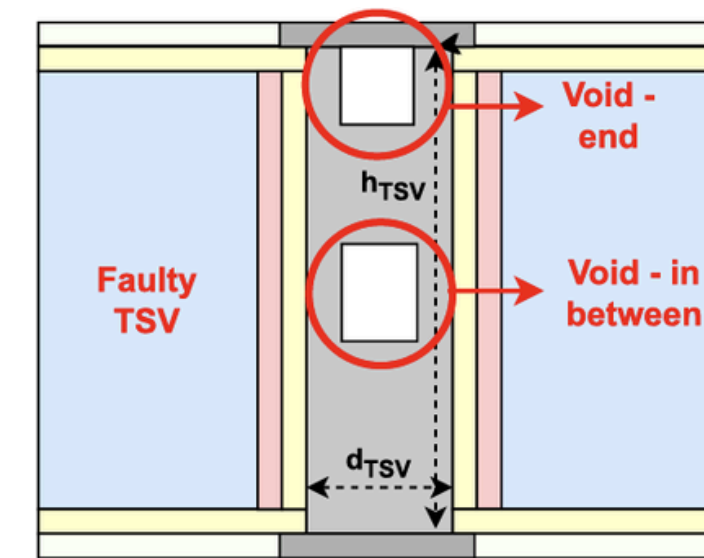


Fig.3.a. Void Defect⁽³⁾

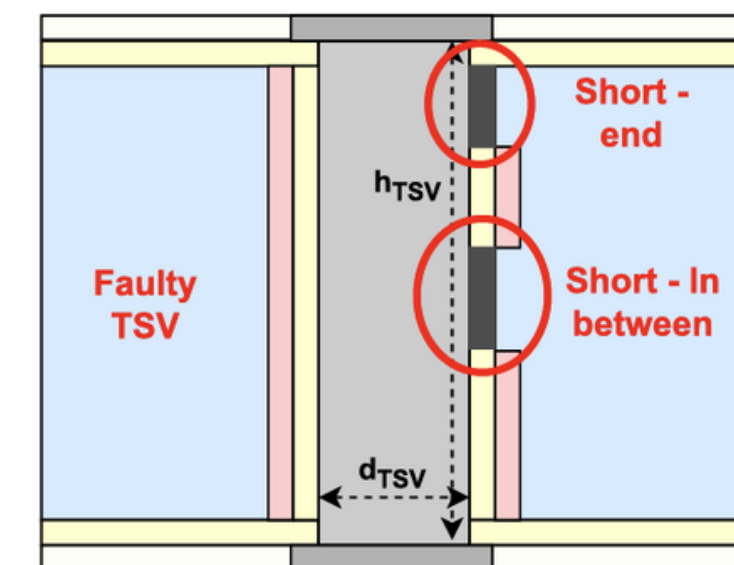
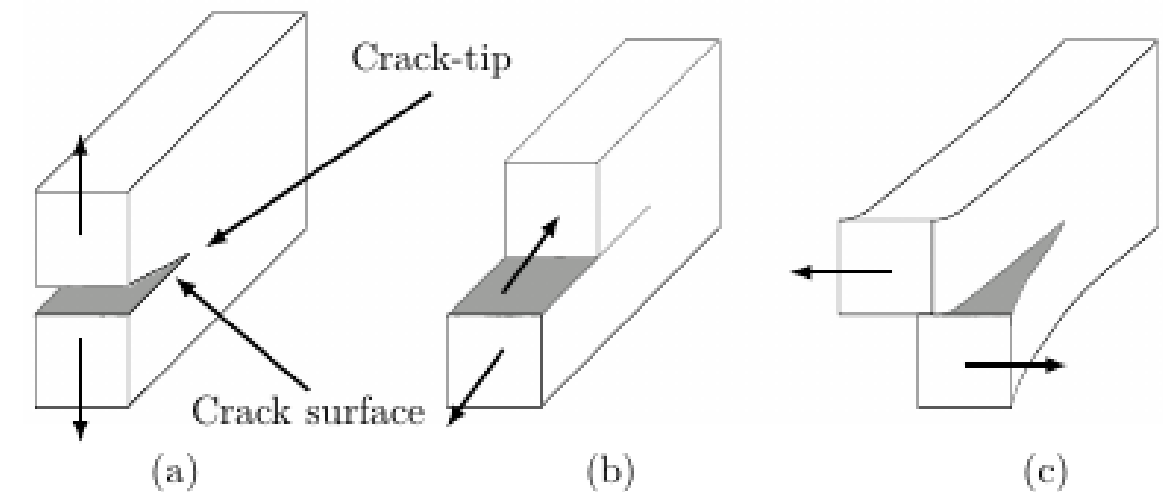


Fig.3.b. PinHole Defect⁽³⁾

Types of Defects

Disconnection Failure: Disconnection failures are caused by the impurities between the TSV's and Bumps

Misalignment: A misalignment defect is the displacement of a TSV from its intended position

Electro-Migration: Electromigration is the movement of metal atoms caused by the flow of high electrical current through a thin conductor.

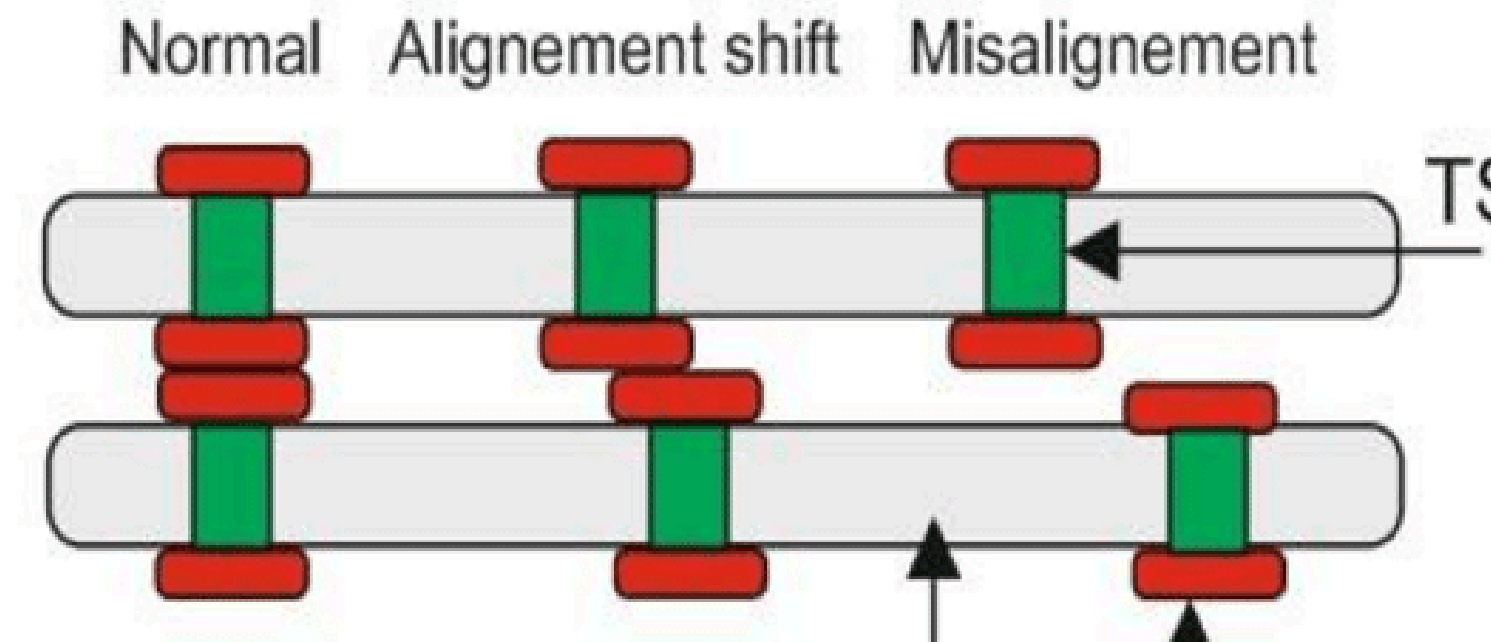


Fig.3.c. Misalignment Defect

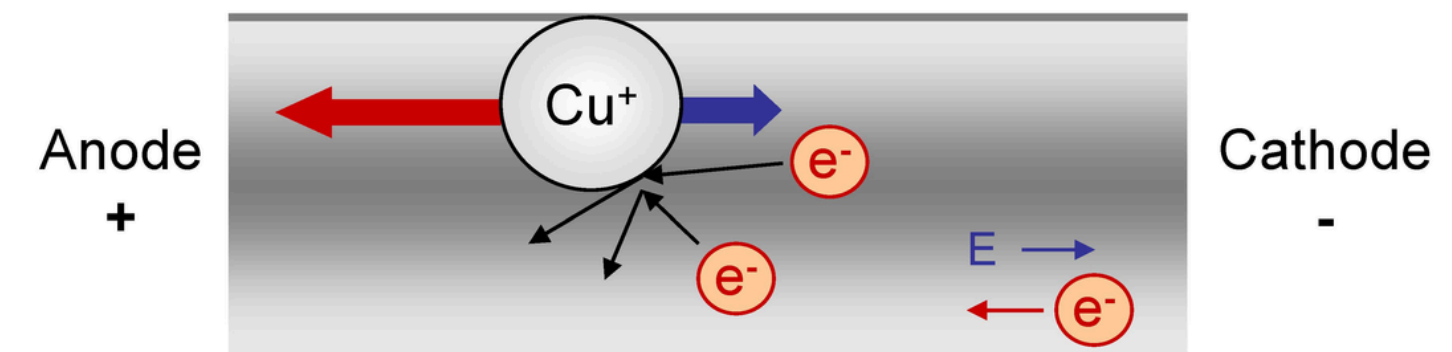


Fig.3.d. ElectroMigration

Focusing PinHole Defect

- Pinhole defects are one of the most common dielectric liner defects in TSVs.
- They create an unintended leakage path between the TSV and silicon substrate.
- They increase leakage current and leakage power.
- They degrade signal integrity by affecting transmission characteristics.
- They can increase noise, crosstalk, and propagation delay.
- They reduce the overall reliability of 3D ICs.

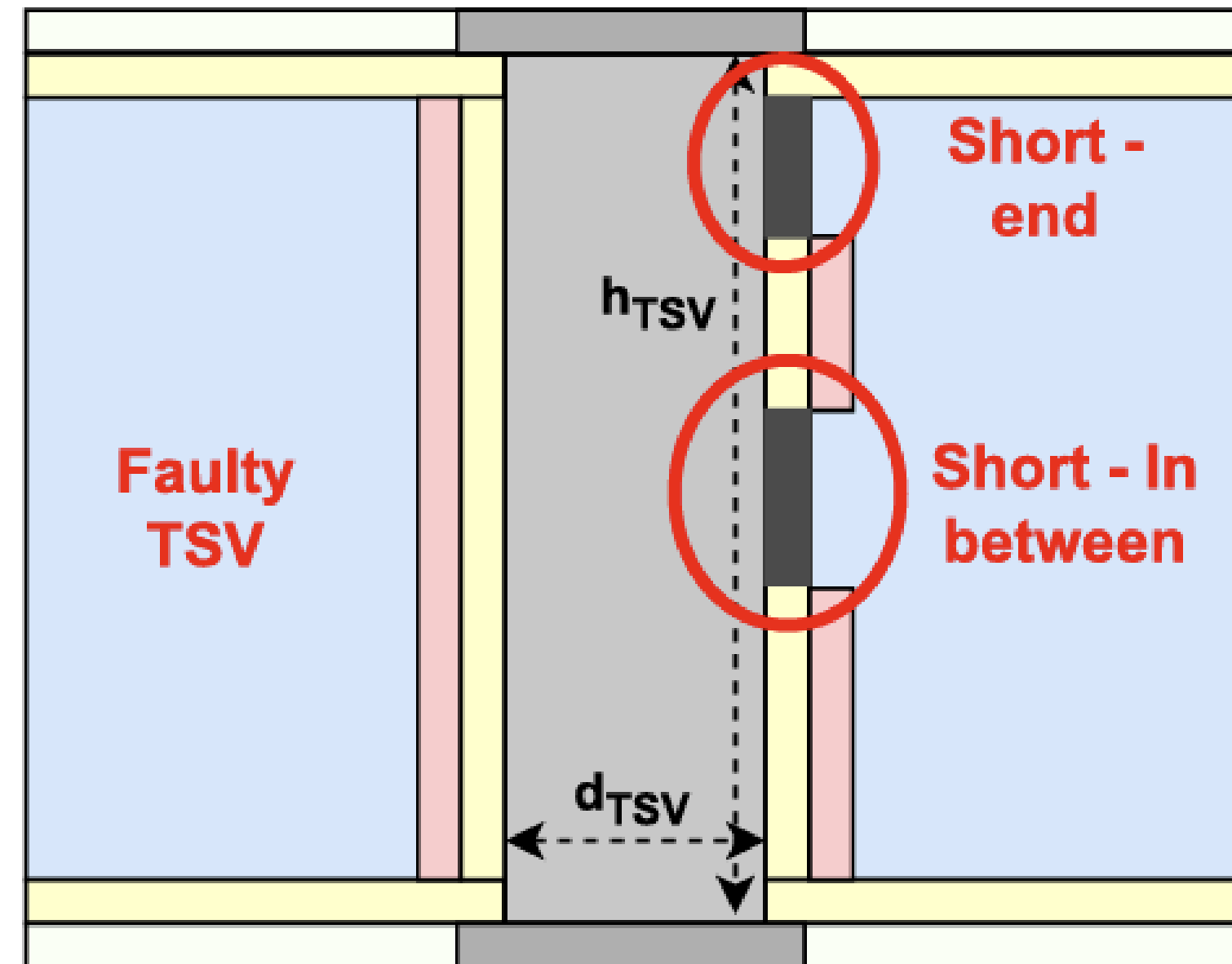


Fig.4. PinHole Defect⁽³⁾

Previous Research Work On The PinHole Defect

Objective:

- Investigated the impact of pinhole defects in TSVs.

Methodology:

- Developed an equivalent RL model for the pinhole defect.
- Analysed the defect using S-parameters (S11, S21).

Findings:

- Evaluated the transmission and reflection characteristics of the TSV.
- Demonstrated that pinhole defects affect signal propagation.

Research Gap:

- The study focused primarily on RL-based modelling.
- Extension to an RLC model may enable more detailed electrical characterization

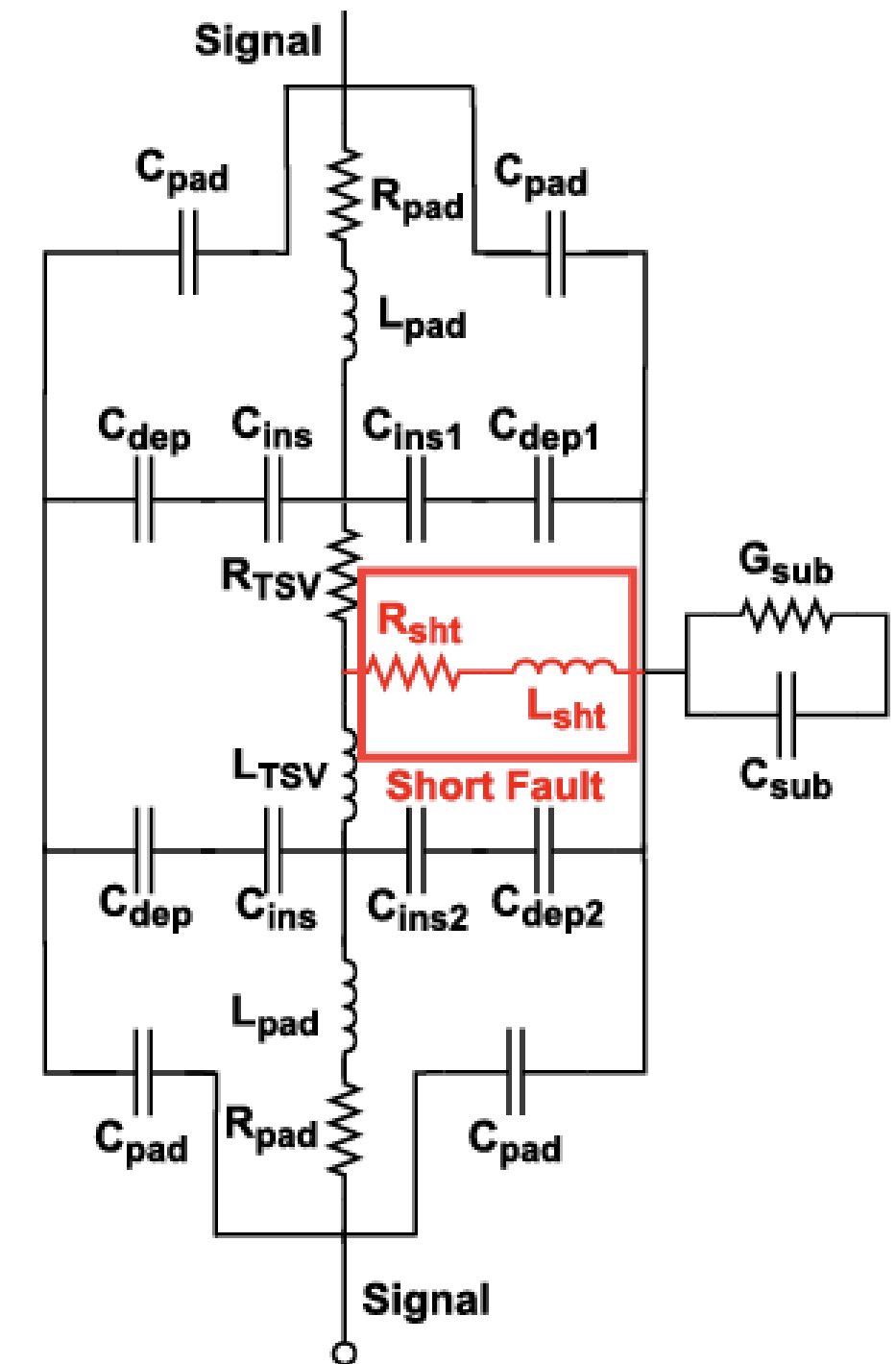
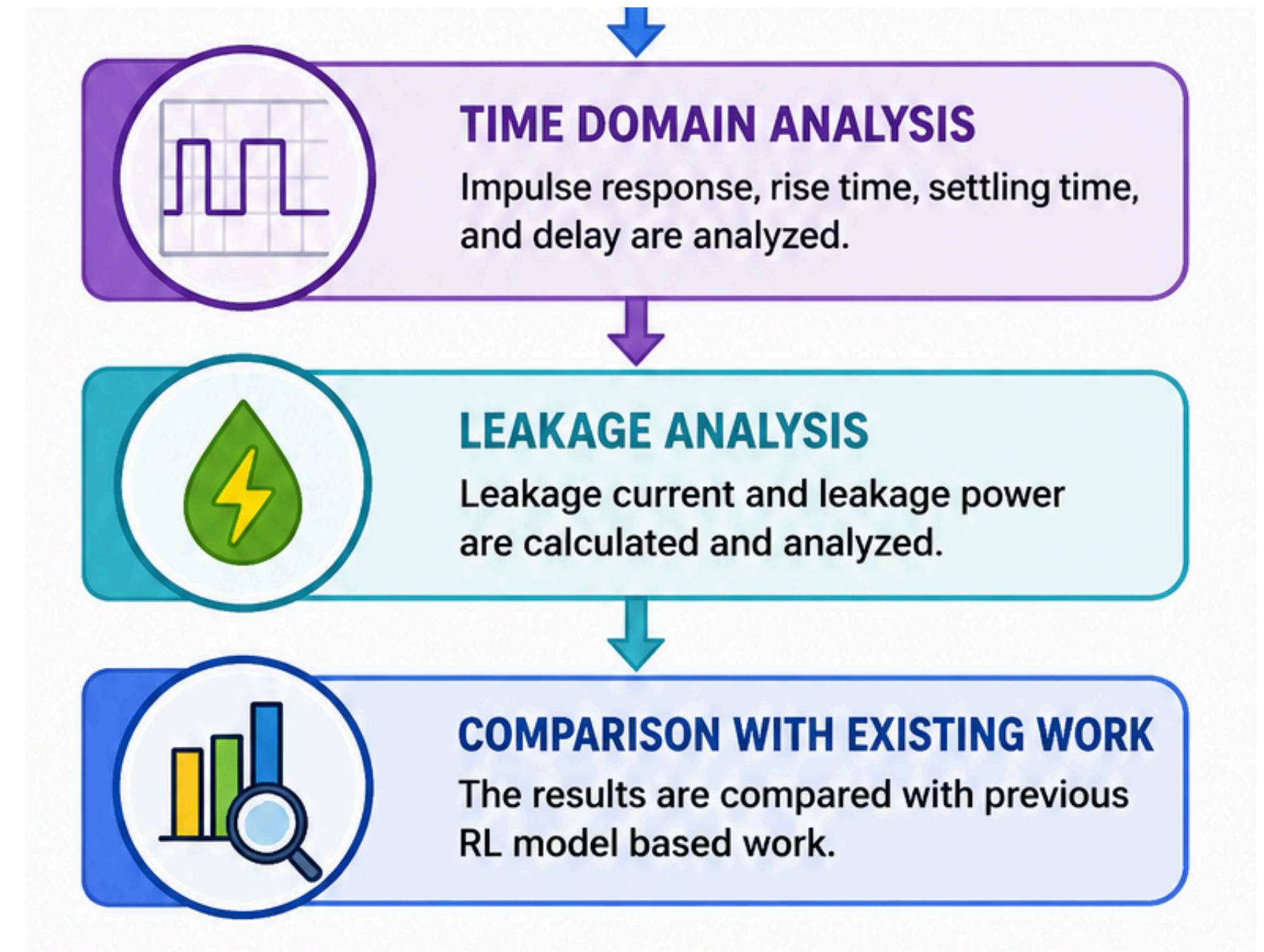
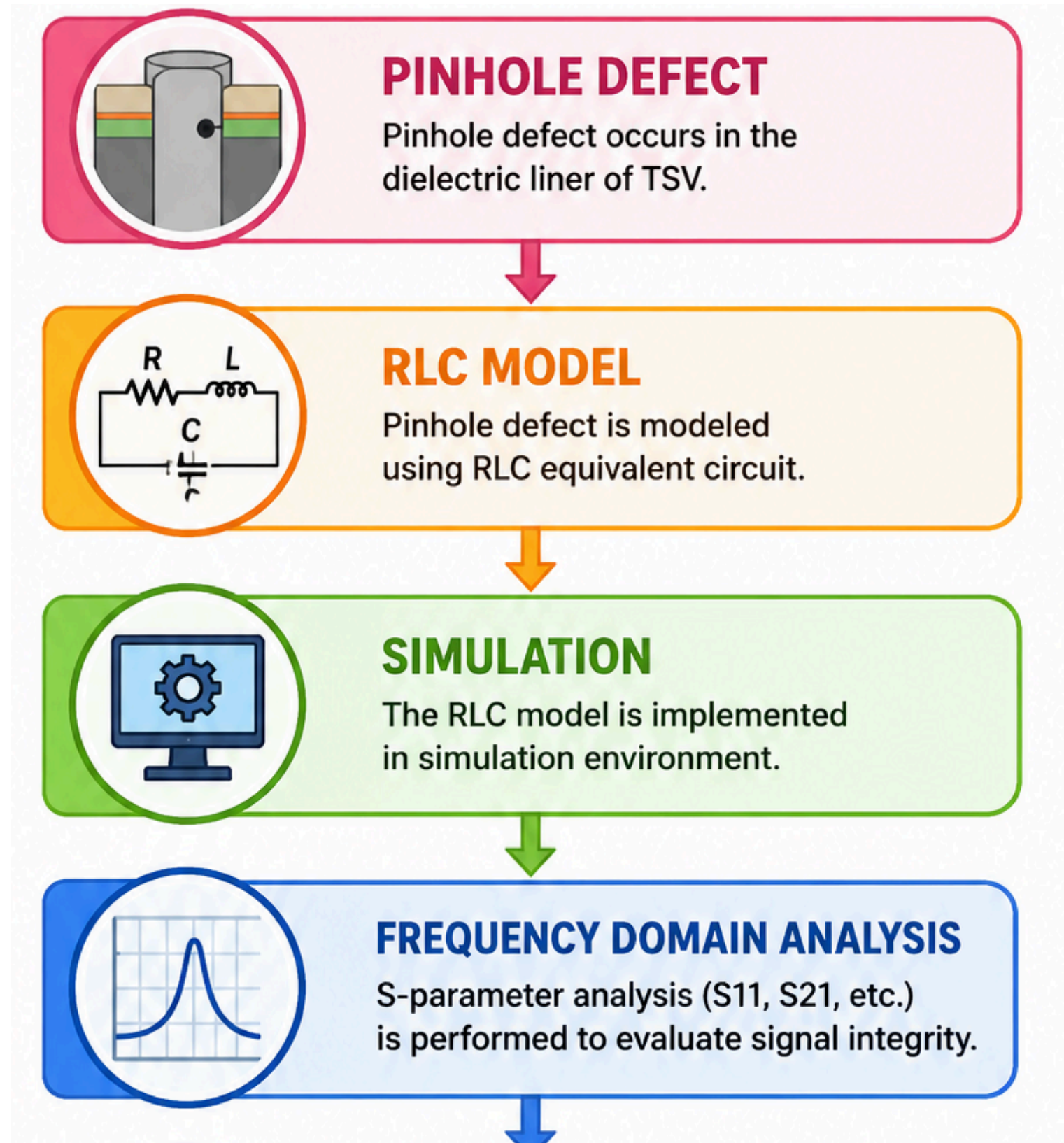


Fig.5. Equivalent circuit of TSV with short fault.⁽³⁾

Proposed Methodology



Parameters



Time Domain Analysis:

1. Noise
2. Crosstalk Delay
3. Leakage Current
4. Leakage Power



Frequency Domain Analysis:

1. $[S_{11}]$ - Reflection Coefficient
2. $[S_{21}]$ - Transmission Coefficient
3. $[S_{13}]$ - Near Field
4. $[S_{14}]$ - Far Field

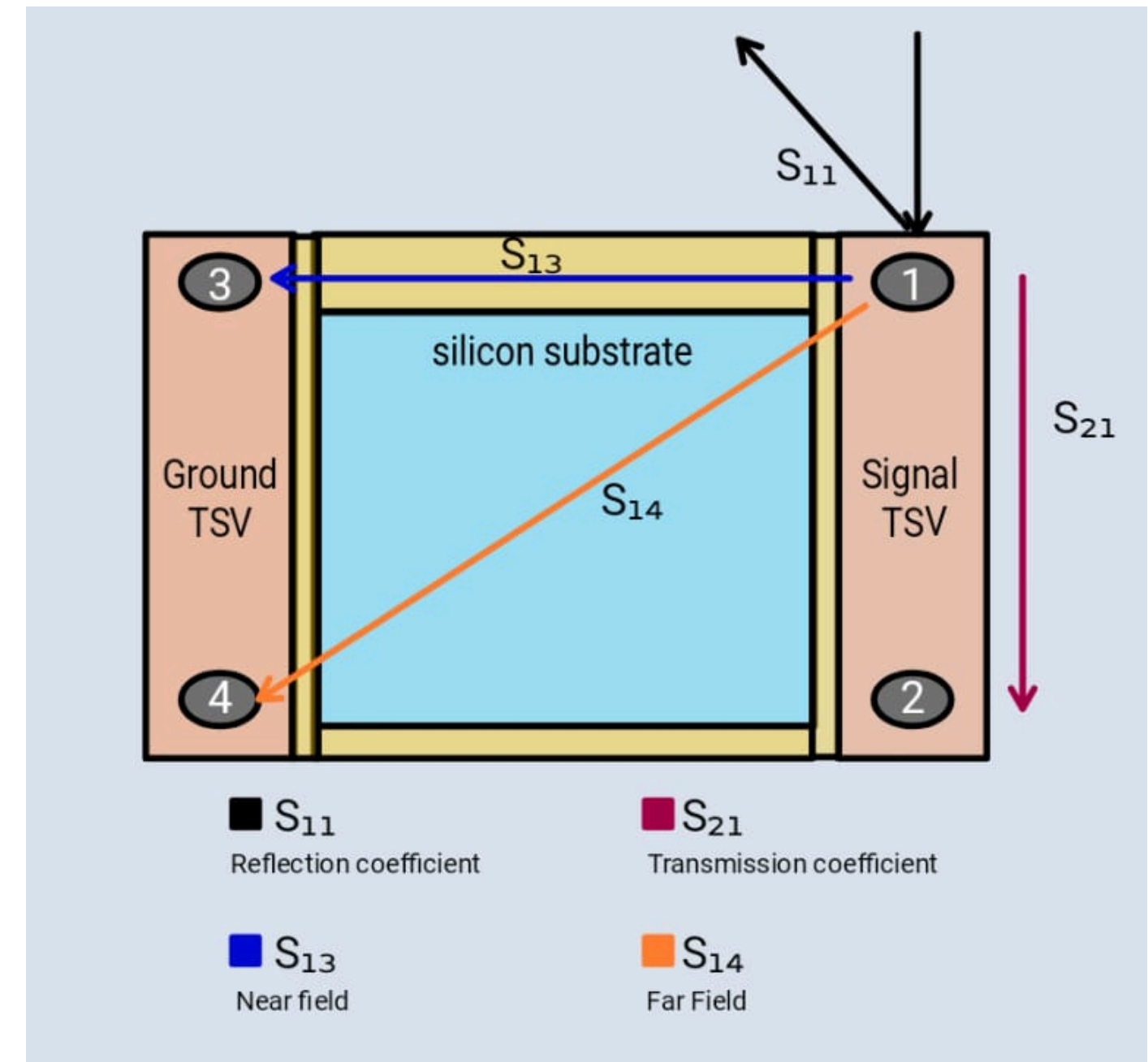


Fig.6. S Parameters

Literature Survey

S.No	Paper	Methodology	Results	Observation
01.	High-Frequency Scalable Electrical Model and Analysis of a Through Silicon Via ⁽¹⁾	Developed a scalable equivalent circuit model using RLGC (Resistance, Inductance, Conductance, and Capacitance) parameters to predict TSV electrical behavior.	The frequency dependent loss of a TSV channel, which is capacitive and resistive, is also analyzed in the time domain by eye-diagram measurements.	The work focuses on TSV electrical modeling and behaviour. Further analysis of TSV defects, such as pinhole defects, and their impact on leakage current, leakage power, noise, crosstalk, and delay can be explored.
02.	Reliability Concerns in Through Silicon Via Based 3D Integration: Fabrication to Packaging ⁽²⁾	Studied TSV reliability challenges from fabrication to packaging	Identified major reliability issues such as thermal stress, electromigration, cracking, voids, and TSV failures.	The electrical impact of individual TSV defects can be further investigated
03.	A Novel TSV Model With Fault Characterization for High-Frequency Transmission in 3D ICs ⁽³⁾	Developed a comprehensive equivalent circuit model for fault-free and faulty TSVs	Demonstrated how TSV faults affect transmission performance and signal integrity at high frequencies.	Additional performance metrics such as leakage current, leakage power, noise, crosstalk, and delay can be further investigated.
04.	Modelling and simulation of TSV considering void and leakage defects. ⁽⁴⁾	This study establishes a unified equivalent circuit model that includes leakage defect TSV, void defect TSV, and defect-free TSV	Demonstrated that void and leakage defects degrade TSV electrical performance and affect signal transmission characteristics.	The study focuses on void and leakage defects. Detailed analysis of pinhole defects, including leakage current, leakage power, crosstalk, noise, and delay, can be further explored.
05.	Prebond Testing of Weak Defects in TSVs ⁽⁵⁾	Proposed testing techniques to detect weak TSV defects before chip bonding. Analyzed defect detection and fault identification methods.	Improved the ability to identify TSV defects during the prebond stage, helping to enhance manufacturing yield and reliability.	Improved the ability to identify TSV defects during the prebond stage, helping to enhance manufacturing yield and reliability.

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